BALLISTIC IMPACT RESPONSE OF THE BASALT/POLYPROPYLENE MESH EPOXY COMPOSITES

A PROJECT REPORT

Submitted by

KUNAM HRUDHAY ROHITH REDDY (0018101156)

SAI SRIKANTH VALLURI (0018101161)

RAVANAM HEMANTH SIVA KUMAR (0018101251)

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HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)
CHENNAI

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HEAD OF THE DEPARTMENT
Dr. R. ASOKAN
Professor& Hod
Department of aeronautical sciences
Hindustan Institute of Technology
and Science, Padur.
Chennai – 603103.

SUPERVISOR
Dr. M. CHANDRASEKAR
Assistant Professor
Department of aeronautical sciences
Hindustan Institute of Technology
and Science, Padur.
Chennai – 603103.

The project viva- voce Examination is held on: _____

Internal Examiner **External Examiner**

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ABSTRACT

The present engineering applications require new and better materials for the replacement of the existing ones. As of now, various kinds of research were going on in the materials industry such as composite materials, especially the laminated composites. When compared to raw or pure or conventional materials, composites have high strength to weight ratio. Fiber reinforced polymer composites are being used in a variety of applications because of their advantages such as ease and various methos of fabrication, relatively low cost of production, superior strength when compared to neat/pure polymer resins. Synthetic fibers such as glass, carbon, basalt etc. have high specific strength but their applications were limited due to higher cost of production. The present research focuses on the characterization and testing of Basalt/Polypropylene mesh epoxy composites with various stacking sequences to study the impact on mechanical properties of these materials. These materials are fabricated and run through various tests to analyze the effects of stacking polymers.

The objective of this study is to investigate and evaluate the ballistic impact response and energy absorbed by the different stacking sequence of pure and hybrid composites.

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CHAPTER 1

INTRODUCTION

1.1. COMPOSITE MATERIALS

A composite material is created by combining two or more materials with different physical and chemical properties. When they're combined, they make a material that's stronger, lighter, and more electrically resistant. They can also help with stiffness and strength. Since they enhance the properties of their base materials and are applicable, they are preferred over conventional materials in many situations.

Composites, also known as Fiber-Reinforced Polymer (FRP) composites, are made up of a polymer matrix reinforced with a synthetic or natural fiber or other reinforcing material. The matrix protects the fibers from external and environmental damage while also transferring the load between them. The fibers, in particular provide resilience and stiffness to the matrix making it more resistant to cracks and fractures.

A composite material's core components are the reinforcement and matrix. Weight and stiffness are provided by the reinforcement. It should be tougher, heavier, and stiffer than the matrix, and it should comfortably bind to it. Typically, the reinforcement is in the form of fibers or particulates.

Composite consist of two substances namely matrix (provides strength and stiffness) and reinforcement (gives protection and load transfer between fibers). Composite materials are classified into many types based on the matrix material. In the broad sense, the term composite materials refer to all solid materials composed of more than one component, wherein those

components are in separate phases.

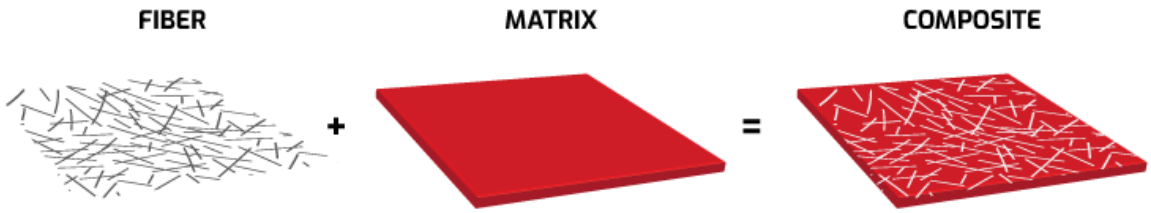

Fig.1.1 Composite

The matrix, on the other hand, serves to bind the fiber content, pass load between the fibers in direction, and shield the fiber from abrasion and the environment. Matrixes come in a variety of forms, including polymers with low strength and stiffness, metals with moderate strength and stiffness but strong ductility, and ceramics with high strength and stiffness but brittleness. They form a close bond between the fiber and matrix in polymer and metal matrix composites, and the matrix transmits loads to the fibers by shear loading at the interface.

Composites are used extensively in a variety of industries, particularly in the automotive and aerospace industries. Composite materials are becoming increasingly common in a number of industries around the world. One of these fields is aerospace, where composite materials have progressed from small use of secondary structures to full use in primary and essential structures in modern airlines. The extended use of composite materials however, has resulted in an increase in composite waste. The demand for fibers and recycled options for used fibers will continue to grow as the global industry develops new uses for composite materials and pushes the limits of what is possible. As a result,

careful handling and recycling of composite materials can be a critical factor for the aerospace industry's continued progression and sustainability.

Fabrication of composites has also gotten simpler in recent years. Fabrication is a manufacturing procedure that involves shaping the resin and reinforcement using some sort of molding. Prior to curing, the mold machinery is required to give the unformed resin/fiber mixture its shape and combined property. Open mold, closed mold, and cast polymer mold are the three styles of composite manufacturing methods. Open molding is a generic term for a method that allows the resin and reinforcement mixture to take form when being exposed to the air. The open mold process has low machine costs, making it suitable for brief production runs. The resin and reinforcement are formed within a vacuum bag or in a closed mold that is not exposed to the environment in the closed mold process.

Hand-layup is the most basic composite fabrication process, which entails inserting layers of dry fabrics, or plies, by hand onto a tool to create a laminate stack. After the layup is over, resin is added to the dry plies. Each ply is covered with resin and debulked after it is put in a wet layup variety. While debulking can be performed by hand using rollers, most fabricators now use a vacuum-bagging process, which entails inserting plastic sheet materials over the layup, sealing it at the tool's edges, attaching one or two ports for air hoses, and then vacuuming the air out of the gap between the sheet and the layup with a vacuum pump. Debulking not only consolidates the layup, but it also eliminates compressed air in the resin matrix, which would otherwise cause unwanted voids (air pockets) in the laminate, weakening the composite. There are a variety of cure options. Allowing cure to occur at room temperature is the simplest process. However, curing can be

facilitated by using heat (typically from an oven) and pressure (via a vacuum). A vacuum bag with breather assemblies is put over the layup and attached to the instrument for the latter, and a vacuum is removed prior to the start of cure. Due to the off-gassing that occurs as the matrix goes through the chemical curing processes, the vacuum bagging process here further consolidates the plies of material and greatly eliminates voids.

1.2. ADVANTAGES OF COMPOSITES

- Composites have high strength to stiffness ratio.
- High impact strength.
- Low weight
- Corrosion resistance
- Low fabrication cost
- The specific strength (strength-to-density ratio) of a composite material is very high. Using these characteristics of composites, airplanes and automobiles move faster and with better fuel efficiency. The specific strength is typically in the range of 3 to 5 times that of steel and aluminum alloys.

1.3. MOTIVATION

Composites are an obvious choice of study in the aircraft industry today. They are used in the majority of structural elements, such as primary structures like wings, fuselages and tails, among others. Despite the fact that composites have improved properties, the processing, development, and recycling costs are high, resulting in an overall rise in the cost of aircraft material construction. The improved mechanical properties of composites

were always desired and always a much appreciated study in materials industry. From the standpoint of a designer, The improvement of composites is critical without sacrificing needed assets. As a result, the project's motivation is to create a more effective composite by stacking layers of composites to improve their mechanical properties.

1.4. OBJECTIVE

The objective of this research is,

To develop the Basalt/polypropylene mesh epoxy composite

- HY-951 Hardener is mixed with LY-556 Epoxy resin.

To determine absorbed energy and the influence of stacking sequence on the energy absorption characteristics of the hybrid composites.

- Ballistic impact response of the Basalt/Polypropylene mesh epoxy composite.

1.5. SCOPE OF THE PROJECT

The project's general goal is to create a reinforced Basalt/Polypropylene mesh epoxy composite, then run a ballistic impact test on it and investigate the amount energy absorbed by the composite and damage failure after the evaluation. Finally, each composite laminate's outcome with various stacking sequence is contrasted.

1.6. SUMMARY

The introduction of composites, motivation of the project, objectives of the project, scope of the project are discussed in this chapter.

CHAPTER 2

LITERATURE SURVEY

2. INTRODUCTION

Kedar S. Pandya, Jayaram R. Pothnis, G. Ravi Kumar, N.K. Naik in 2013 have conducted a study on Ballistic impact behaviour of hybrid composites. The ballistic impact behaviour of different polymer matrix composites has been discussed in the paper. Quantitative data for ballistic impact behaviour of six types of polymer matrix composites was acquired.

Four types of hybrid composites were made using 8H satin weave T300 carbon fabric and plain weave E-glass fabric with epoxy resin. The projectile used for the impact is a right circular cylinder of hardened steel with a mass of 4.15 gm, length of 8.82 mm and diameter of 8.74mm.

The hybrid specimen were made through matched die molding and the glass and carbon layers were stacked alternately for each hybrid.

The experimental setup is arranged as such that a hollow shield is used from the tip of the barrel up to the impact chamber to protect the user of the test apparatus from accidental stray firings and the center lines of the barrel, hollow shield, chronograph and the center of the specimen after clamping were aligned.

The observations used for the comparison were the ballistic limit velocity and the energy absorbed per unit areal density.

It is observed that the order of ballistic limit velocity per unit areal density of hybrids is different to the hierarchy of energy absorbed per unit areal density of the hybrids.

The damage patterns, intensity of strike and dimensions of each crack of each of the specimen were observed after the impact and were documented for further comparison of the specimen.

Elias Randjbaran, Rizal Zahari, Nawal Aswan Abdul Jalil, and Dayang Laila

Abang Abdul Majid in 2014 have presented a research article on Hybrid composite laminates reinforced with Kevlar/Carbon/Glass woven fabrics for ballistic impact testing. The effects of stacking sequence layers of hybrid composite materials on ballistic energy absorption is studied throughout the current study.

The five hybrids considered for the study each consist of two layers each of Kevlar, glass and carbon in different stacking sequence to observe the effects of different stacking sequences on the energy absorption of the hybrids.

The hybrid composite construction was made by hand lay-up method with 50 parts EPOKUKDO YD -128 epoxy resin and 50 parts Polyamide-Domide (A.V:350) hardener by weight which were comprehensively cured after seven days at 20°C.

A cylindrical AISI4340 steel pellet was used as the projectile during the impact testing of the specimen.

The observations made during the test were ballistic energy absorption, final velocity after impact, perimeter of the impact zone on the specimen and the energy absorption component was highest in the hybrids compared to the pure carbon specimen.

Stephen C, Shivamurthy B, Mourad A.H.I. and Selvam R in 2021 have made a detailed study on High velocity impact behavior of hybrid fiber-reinforced epoxy composites. The tensile, flexural and hardness characteristics were investigated during the study.

The pure laminates as well as hybrid composite laminates of Kevlar, Carbon and Glass with various stacking sequences were hand lay-up process which was then followed by hot compression molding technique. The curing process is carried out at 100°c with 13.85Mpa for 30 minutes in compression molding.

The specimen were put through tensile test to determine the tensile strength and tensile modulus, burn-off test to determine the vol% of fiber, flexural test to

determine the flexural behavior, Rockwell-B hardness tester to determine the laminate hardness and the laminate fracture is observed under scanning electron microscope.

The observations made during the tests lead to the conclusion that hybridization of reinforced fabrics improves the stiffness of laminates for ballistic applications and leads to cost reduction by sandwiching glass fabric inside Kevlar composites with their mechanical properties intact.

Pol M.H. and Liaghat G have made an investigation of high velocity impact behaviour of nanocomposites in 2016. The ballistic behaviour of glass/epoxy/nano clay nanocomposites was studied during this investigation.

The resin is prepared by adding nanoparticles with hardener for 60 min at 3000 rpm and sonicated for 30 min with 150 kW/cm^2.The mixture is again stirred for 60 min at 3000 rpm. The mixture is then stirred with Epon 828 resin for 10 min at 800 rpm at 25°C.

The laminates have 12 layers and 50% fiber volume fraction and were made using lay-up method up to thickness of 2.3mm and were cured for 150min at 80°C and 2.5 bar pressure followed by 150min at 120°C and 4 bar pressure in an autoclave.

The projectile used for the ballistic impact test was a flat-ended steel tip of length 15 mm, mass 8.9 g, 10 mm diameter and 134 m/s and 169 m/s impact velocities. The initial and the residual velocities were recorded for each test specimen for further observation of improved characteristics of the nanocomposites.

The observations made after the test were that the ballistic impact resistance is increased for glass/epoxy laminates by adding nano clay particles. The hybrid nanocomposites show an increased energy absorption capability of 10% with 3 wt.% of nano clay content. The best energy absorption capability was observed at 10% nano clay content with 169 m/s ballistic speed and absorption was 20%.

Pekbey Y, Aslantaş K and Yumak N in 2017 have made a detailed study on Ballistic impact response of Kevlar Composites with filled epoxy matrix. The ballistic impact behaviour of Kevlar with nano clay, nano calcite and nanocarbon as fillers were studied throughout the experiment.

The specimen were made by hand lay-up process with nano clay and nano calcite as fillers with 1 wt.% and 2 of epoxy resin hardener mixture whereas nanocarbon were dispersed into the epoxy in 0.5%, 1% and 2% in weight ratio to the epoxy matrix. Ultrasonic cavitation technique is used to disperse nanoparticles in the polymer matrix and the laminate specimen contain 20 layers each.

The observations made during the study were termed based on damage and perforation.

The addition of nano clay and nano calcite to ballistic materials to improve the energy absorption capabilities at lower weight is already a wide spread strategy utilized in materials industry.

It was observed during the study that the addition of nano clay and nano calcite have brought with them a drastic improvement in ballistic performance. But after a certain weight ratio the increased addition of nanoparticles leads to the decreased ballistic performance. The laminates with 1 wt.% of nano clay as filler for epoxy resin give the best result in performance improvements. The addition of nanocarbon with Kevlar/epoxy laminates leads to decreased ballistic impact resistance and failure mechanism.

Clifton Stephen, B.H.S. Thimmappa, Rajiv Selvam, B. Shivamurthy in 2019 have studied the polymer nanocomposites for high-velocity impact applications. The specimen taken for observation were Carbon, Kevlar, Glass and hybrid fabrics reinforced with suitable nanophase materials.

Epoxy matrix such as polypropylene, High-density polyethylene and polycarbonate were used as for nanofillers nanoparticles such as Bucky paper, carbon nanotubes, graphene platelet nano powder, silica nanoparticles, nano clay,

milled nanofibers and nano- Al_2O_3 were used for reinforcing the polymers.
The fabrication method preferred for the process was Vacuum-assisted resin transfer molding. The reinforcement materials were infused into the matrix by applying vacuum ideal for low-cost fabrication. The researchers who study HVIR laminates widely prefer this method.

The molecular dynamics (MD) simulations and dissipative particle dynamics (DPD) simulations were used for analysis of each of the designed samples for ballistic impact performance improvements.
The analysis and all the previous studies reveal that the hybrid nanocomposites have the potential to bring composites to the next level of structural and high-velocity impact resistance applications.
This study also reveals the best application possible for each of the hybrid nanocomposites used in analysis in various sectors currently in use.

Mahdieh M. Zamani, Abdolhossein Fereidoon, Alireza Sabet have made a detailed analysis on high-velocity impact response and mechanical properties of multi-walled carbon nanotube (MWNT)-filled polypropylene(PP) nanocomposites in 2012.
The analysis was made on polymer nanocomposites containing 0.75, 1.0, 1.5 wt.% of multi-walled carbon nanotubes in a polypropylene matrix in relation to low and high velocity performances.
Prior to the preparation of nanocomposites, polypropylene and the MWNT were vacuum dried in an oven at 80°C for 48 hours. The mixing was carried out by an internal mixer with electrically heated mixing head and a pair of high shear roller type rotors. The mixing conditions were 180°C and 120 rpm blended for 10 min. The specimen were then molded into dog-bone shape. The nanocomposites were made by melt compounding and injection molding process.
The morphology of nanocomposites is observed through a scanning electron

microscope and a transmission electron microscope is used to investigate the morphology of the MWNT.

The observations made after the test results were that Izod impact strength increases for the notched specimen of nanocomposites and maximum ballistic limit velocity, energy absorption and the minimum decrease in residual velocity were observed in polymer nanocomposites with 1 wt.% MWNT with 100% increase in energy absorption potential and the young's modulus and yield strength also increases.

The ductility of the nanocomposites decreases with increase in the MWNT content. But the maximum increase was observed for nanocomposites with 1wt.% MWNT. Flexural modulus and flexural strength of MWNT/PP nanocomposites increases with addition of CNT in polymer matrix.

CHAPTER 3

METHODOLOGY

3. INTRODUCTION

Methodology is the process of material selection and the process of fabrication along with the laminate cutting.

3.1 MATERIAL SELECTION

3.1.1 INTRODUCTION

The most important process in engineering design and fabrication of a component is material selection and collection. The materials chosen must meet the requirements of the desired output. It should be able to resolve the issues and challenges that are being tackled.

3.1.2 REINFORCEMENT

Reinforcement can be fibers, fiber particles or whiskers (nano or micro) particles, these reinforcements are generally used to increase the mechanical properties of the composite. Strength and stiffness of a composite is provided by reinforcement.

3.1.3 BASALT FIBER

Basalt is a natural fiber that is easy to extract by one stage process homogenization and extraction of fibers. It is cheaper than carbon and has better properties than glass fiber so it is used as a fiber reinforcement material which reduces the cost of the composite material and develops new material for various purposes and applications such as aerospace, automotive and fireproof textiles. It consists of olivine, minerals plagioclase and pyroxene. Woven fabric is highly

useful in construction industry as asbestos substitute is present for their better specific strength i.e., three times greater than steel.

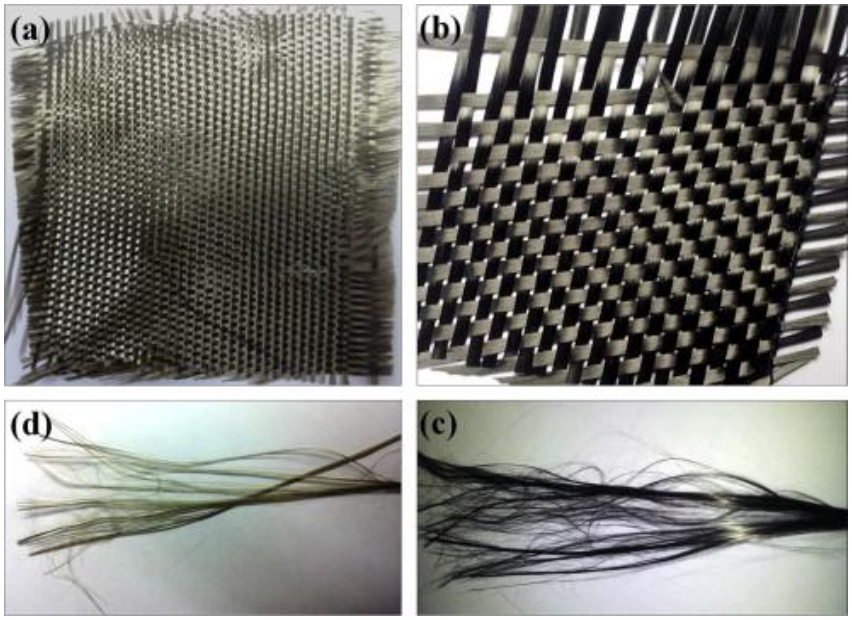

Fig.3.1 Basalt Fiber

3.1.4 MANUFACTURING OF BASALT FIBER

Naturally occurring basalt rock is fed into the weighing chamber and crushed mixture enters the furnace. It is liquefied at 1500°c. From taking over the fibers and sizing the fibers filaments drawn and stored. Basalt differs from glass by being opaque and transparent in nature absorbs more energy; hence maintain heat over basalt mix is tedious. These fibers have flexibility by chopping and using as reinforcement in matrix of composites. Continuous fibers yarns useful in woven fabrics for textile industry.

3.1.5 ADVANTAGES OF BASALT FIBER

- Good fire resistance.
- Chemically inactive to environments.
- Acoustic and vibrational insulation.
- They work under high temperature conditions.
- Basalt fiber has 15-20% of high tensile strength and modulus.

3.1.6 POLYPROPYLENE MESH

Polypropylene mesh is synthetic mesh, it has less tension and not more flexible nature. It has more uses in medical field rather than in engineering. It has uses in hernia and soft tissue repairs.

Fig.3.2 Polypropylene Mesh

3.1.7 MANUFACTURING OF POLYPROPYLENE MESH

Polypropylene mesh is made up of most widely used plastic in market. Polypropylene is made up of polymerisation of propylene and then the fibers were made out the product. These fibers were woven into fabric for making mesh.

3.1.8 ADVANTAGES POLYPROPYLENE MESH

- Longer life or more durability.
- Available in larger sheets.
- It will expand more to create larger volumes.
- Rigid in nature.
- Excellent elasticity.

3.1.9 MATRIX

Resin is used to transfer the stress between the reinforcing fibers of a composite. Resin act as a glue to hold the fibers together and protect the fibers from mechanical and environmental damage.

3.1.10 EPOXY RESIN

Epoxy resin is used to bind the fibers together and distribute the loads between the fibers. In general low yielding resin is mixed with high yielding hardener to enhance it properties and to increase its strength. Epoxy resin is used for adhesive purpose, commonly it is used in the manufacturing of the composites for aircrafts etc…

Fig.3.3 LY-556 Epoxy Resin.

3.1.11 RESIN SELECTION

Because of its compatibility with Basalt and commercial availability, epoxy resin (Araldite LY556) was chosen.

Epoxy has the following characteristics:

- Increased mechanical strength and better fatigue strength.
- Excellent adhesion to various substrates.
- Excellent electrical insulation.
- Low shrinkage during curing process.
- Excellent moisture resistance.
- Low tendency to crystallize.
- Excellent chemical resistance.
- Long shelf life.

3.1.12 HARDENER SELECTION

Epoxy resins are only cured when a curing agent, such as a hardener, is added. The curing agent Araldite HY951 is used. Araldite HY951 is an aliphatic primary amine that cures at room temperature or in a shorter amount of time at higher temperatures. HY951 was chosen due to its ease of availability and low cost.

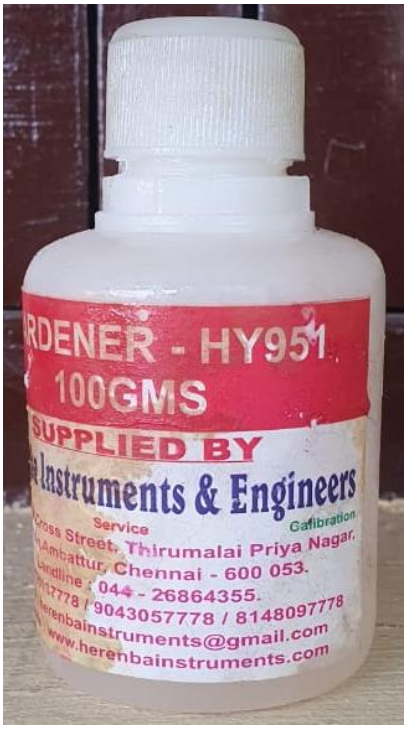

Fig.3.4 HY-951 Hardener.

3.1.13 SUMMARY

The selection of fiber, mesh, resin and hardener is discussed along with their manufacturing process and advantages are discussed in the material selection.

3.2 FABRICATION

3.2.1 INTRODUCTION

Fabrication is the process of molding fiber and matrix together and cure them to form composite.

Fabrication techniques mainly depends upon three factors

- The properties of matrix and reinforcements.
- Shapes, sizes and engineering standards.
- Future applications.

The techniques of fabrication vary from one commodity to the next. There are two types of manufacturing methods used in the manufacture of composites: open-mold and closed-mold processes.

3.2.2 OPEN-MOLD PROCESS

Open molding process is typically used for the most of all composite fabrication methods, part size and design choices are nearly unlimited. The open molding process is typically used for a wide variety of items that cannot be manufactured in more automated processes, or for low-volume parts that cannot justify the higher mold costs of automated processes.

Open-mold process have following methods:

- Spray lay-up
- Hand lay-up
- Filament winding

SPRAY LAY-UP:

Spray lay-up is an open molding composites fabrication technique that involves spraying resin and reinforcements onto a reusable mold. The resin and glass may be applied separately or chopped together in a single laminate. After that secondary spray-up layer embeds the centre between the laminates, which can be made of wood, foam, or another material. After that, the component is cured, cooled, and then removed from the mold.

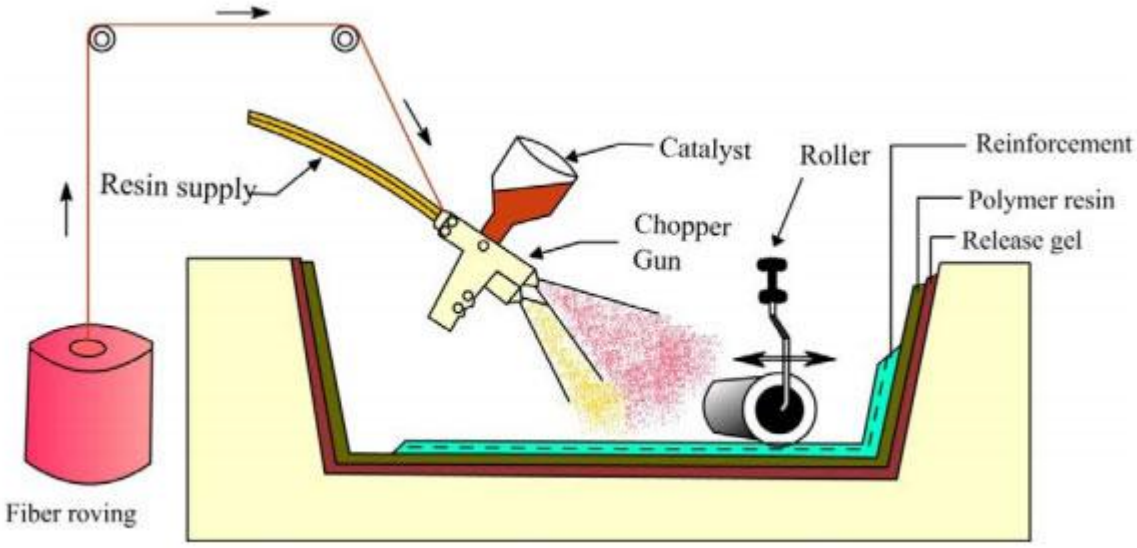

Fig.3.5 Spray Lay-Up Technique

FILAMENT WINDING:

Filament winding is a fabrication method that is primarily used to create open (cylinders) or closed end structures (pressure vessels or tanks). The filaments are wound under tension over a rotating mandrel in this horizontally in line with the axis of the spinning mandrel, laying down fibers in the desired pattern or angle, while the mandrel rotates around the spindle.

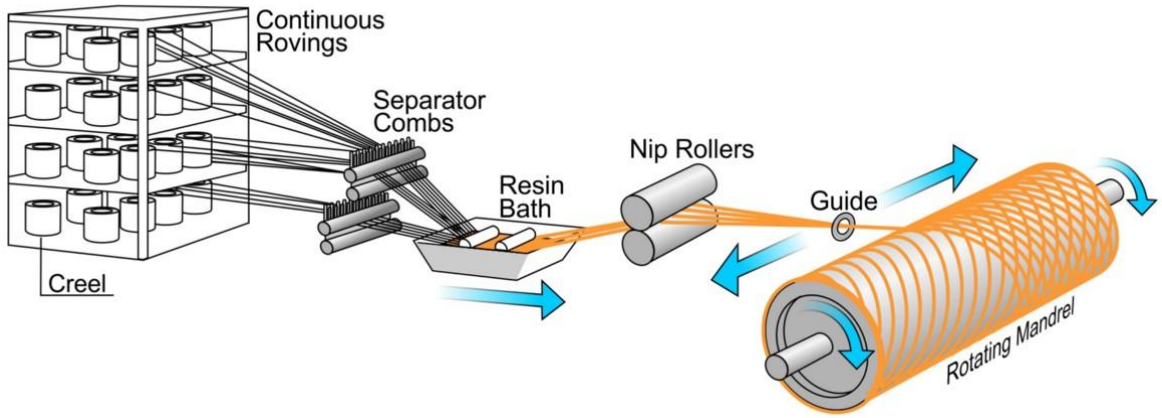

Fig.3.6 Filament Winding Technique

HAND LAY-UP:

Hand lay-up method is the oldest and most widely used technique of all. Layers of unidirectional or bidirectional fibers are distributed in one or more ways, then resins are applied using rollers or brushes to evenly distribute the resin across the fibers. After that, the wetted, stacked layer of fibers ia allowed to cure in the open air.

Hand lay-up technique can be employed for:

- Obtaining sandwich constructions.
- Less tooling costs.
- Large and complex parts can be produced.
- Higher fiber content and longer fiber than spray lay-up,

The disadvantage of this method is that it produces only one molded surface and is a low volume operation, Curing can take a long time in some cases. The viscosity of the resin should be low for fast use. Even, if the efficiency of the labour working is low, there might be further wastage.

This is commonly used for basic wing rotor blades, production vessels, and other similar applications.

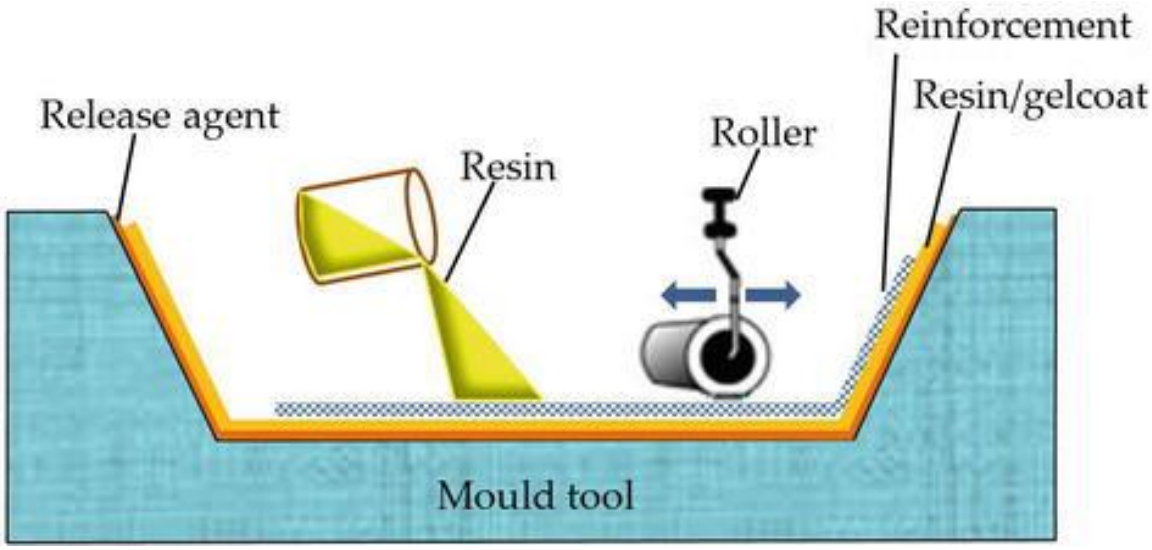

Fig.3.7 Hand Lay-Up Technique

3.2.3 CLOSED-MOLD PROCESS

This technique is used on buildings that are sealed off from the outside world. Closed molding processes, as opposed to open molding, enable manufacturers to produce higher-quality elements more quickly and more systematically, with less waste. The surface cosmetics of finished elements are higher, eliminating the need for post-processing.

Closed mold processes have the following methods:

- Compression molding.
- Vacuum bag molding.
- Resin transfer molding.
- Injection molding.

COMPRESSION MOLDING:

It is the technique that involves in placing a feed into the heated mold and

then sealed with top plug, squeezed with high hydraulic (pressure) press to make sure all the material stick to each other with no air gaps and also excess resin will be extracted through outlets or ejector pin and then sealed air tight. In the hot mold the material gets cured.

The following are some advantages of compression molding:

- Air molecules or voids can be avoided in this process due to high pressure applied.
- Large number of parts can be produced due to its compatibility, but the tooling cost is more and the preparation of mold would be high cost.

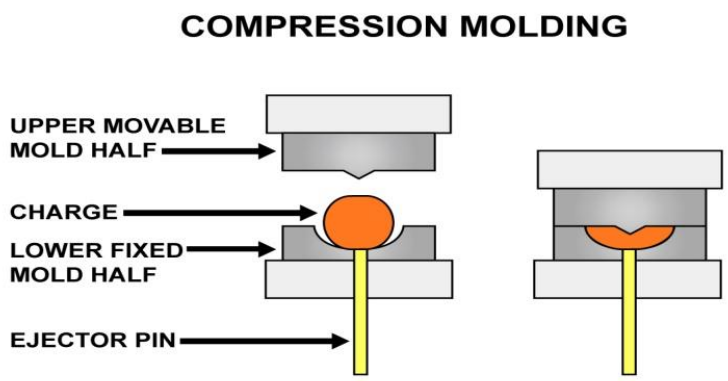

Fig.3.8 Compression Molding Technique

VACUUM BAG MOLDING:

It was the modification of open mold process such as hand lay-up and spray lay-up. Layers of fiber and resin were placed alternatively by hand in to the mold and then inserted in to vacuum bag and by reducing the pressure inside the bag less than atmospheric pressure. Due pressure differences inside and outside of the vacuum bag the external atmospheric pressure acts as load on the laminate and removes the trapped air, excess resin and compacts the laminate.

The following are some advantages of vacuum bag molding:

- Excess resin can be removed from this technique when compared to hand lay-up and spray lay-up techniques.
- Bonding between layers can be improved and complete fiber wet out can be archived.

Vacuum bag molding technique requires higher level of skill by operators and it adds an extra cost for labour and for dispose of bagging materials.

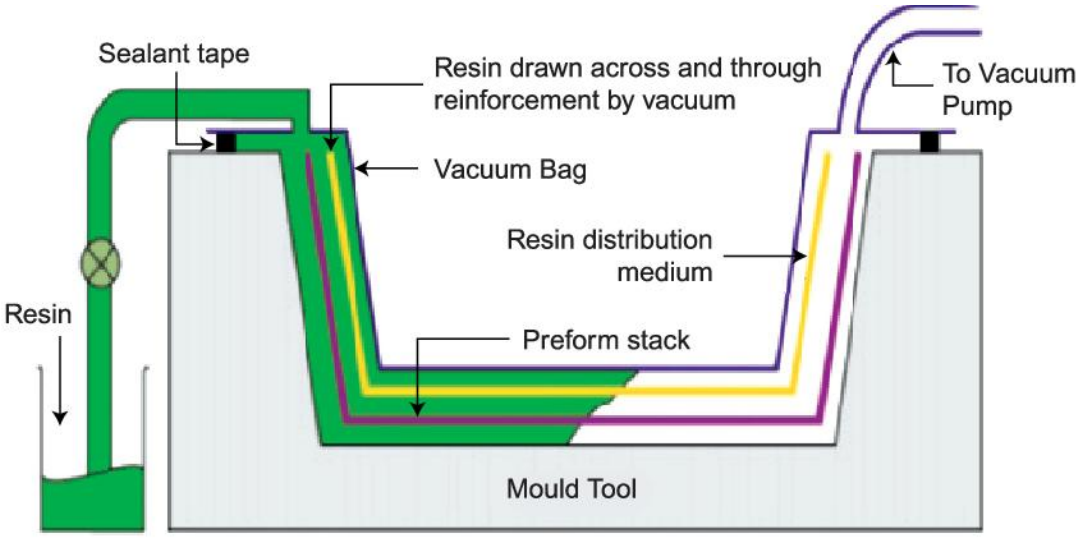

Fig.3.9 Vacuum Bag Molding Technique

RESIN TRANSFER MOLDING:

It is an intermediate volume molding process for producing composite laminates. Gel coats are applied for high quality durable finish. Reinforcements or fibers are placed into mold in the required shape under pressure. The mold cavity allows precise thickness for a smooth finish.

The resin is then pumped into the heated mold under pressure, displacing air through vents until the mold is filled with resin then the composite will be cured depends upon the mold temperature.

The following are some advantages of resin transfer molding:

- Low environment impact.

- Short production cycle.
- Good surface quality with zero trap of air in the product.

The tooling costs can be expensive due to complex mold designs and special mold maintenance is required as the inserts have to be taken out.

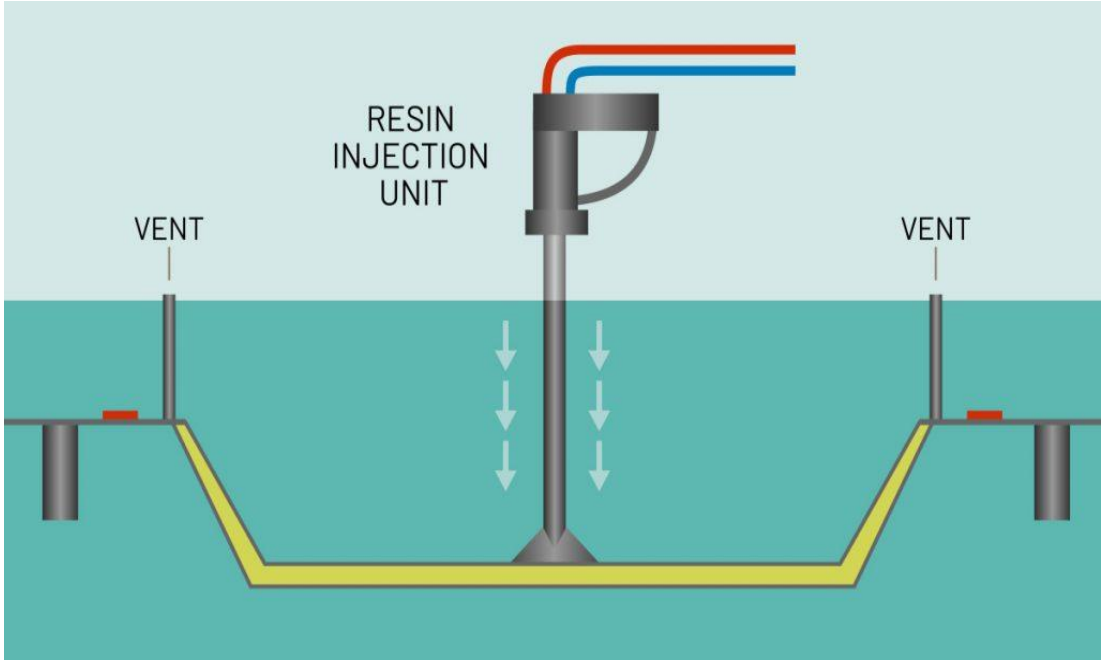

Fig.3.10 Resin Transfer Molding Technique

3.2.4 FABRICATION PROCESS

We have used hand lay-up technique to fabricate composite material.

Its fabrication involved several stages:

- Cutting Basalt and PP Mesh in to required dimensions.
- Mixing of Hardener HY-951 with Epoxy Resin LY-556.
- Stacking up the layers in requires manner with resin mixture in between each layer of reinforcement, followed by adding dead weights on top of the mold to squeeze excess resin.

CUTTING OF BASALT/POLYPROPYLENE MESH AND MIXING OF EPOXY WITH HARDENER:

Eighteen pieces of Basalt fiber and seventeen pieces of polypropylene mesh were cut into required dimensions (300mm * 300mm).

Epoxy was mixed with hardener for each laminate. The ratio of reinforcement to epoxy was taken as 1:3 and the hardener will help the material to cure faster.

STACKING OF LAYERS FOLLOWED BY MOLDING PROCESS:

Mylar sheet was used as the base mold, wax was applied on the mylar sheet as releasing agent to make it easy to separate laminate from the sheet. The silicon rubber frame was set and first layer of epoxy resin and hardener mix layer was spread over the wax coated mylar sheet. The first layer of reinforcement was placed over and another layer of resin and hardener mix layer was spread again, this process continues till our desired number of layers got completed. Hand lay-up with a roller is used for this stacking operation. Another layer of mylar sheet was used on top of all the stacking. Dead weights were placed over the top layer of mylar sheet for curing and bonding purpose.

After the curing, processed laminate is taken out. The mylar sheets were removed from the laminate for the further process.

STACKING SEQUENCE:

Table-3.2.4 Stacking Sequence

Sample-1 $[P]_5$	Sample-2 $[P]_2/[B]/[P]_2$	Sample-3 $[B]/[P]_3/[B]$	Sample-4 $[P]/[B]_3/[P]$	Sample-5 [B]/[P]/[B]/[P]/[B]	Sample-6 $[B]_5$
PP MESH	PP MESH	BASALT	PP MESH	BASALT	BASALT
PP MESH	PP MESH	PP MESH	BASALT	PP MESH	BASALT
PP MESH	BASALT	PP MESH	BASALT	BASALT	BASALT
PP MESH	PP MESH	PP MESH	BASALT	PP MESH	BASALT
PP MESH	PP MESH	BASALT	PP MESH	BASALT	BASALT

Fig.3.11 Fabrication step by step process

CUTTING OF COMPOSITE LAMINATE INTO SPECIMENS BY USING MECHANICAL BAND-SAW MACHINE:

Markings are made on the composite laminate according to required dimensions of the specimens that are to be tested. Then the laminates were carefully taken to the machine by wearing gloves to avoid minor injuries. Blade runs at high speed or rpm and specimen was introduced slowly and carefully so that the specimen will be cut neatly and we can get the required shape and dimensions.

Fig.3.12 Vertical Band Saw Machine

Fig.3.13 Basalt Fiber Composite

Fig.3.14 PP Mesh composite

Fig.3.15 Basalt/PP Mesh Hybrid composite

3.2.5 SUMMARY

Introduction of fabrication process, different kinds of fabrication process, cutting of the reinforcement in to required dimensions, stacking sequence and cutting of laminates are discussed in this chapter.

CHAPTER 4

TESTING

4. INTRODUCTION

In this chapter we are going to discuss about the Ballistic impact testing and amount of energy absorbed with its conclusion and future scope.

4.1 BALLISTIC IMPACT ANALYSIS

Ballistic impact analysis the process of sudden impact of small spherical weight (bullet) on the composite material and calculating the amount of energy absorbed and the maximum loads it can with stand. Generally, this method is used to test the armours such as vests, armed vehicles that store projectiles.
In low velocity Ballistic impact testing, the bullet or small mass moves at 100-above m/s. This method is more preferred for testing impact resistance of composite materials.

The Ballistic impact testing setup consists of a high air compressed gun through which a small spherical mass or bullet is projected on to the testing sample, solenoid valve is placed in between barrel and high air compressed gas chamber which is used to trigger the bullet by suddenly releasing air at high pressure. The barrel is 1 meter long and 3 cm in diameter which is coated with plastic for smooth movement of the bullet inside the barrel. The plate is fixed in the target holder. The target holder designed to hold material of 200 mm square and 300 mm square and the material thickness of 1 mm to 15 mm. The shot from machine impact, the plate is fixed to the holder.

Fig.4.1 Ballistic Impact Setup

4.2 SPHERICAL BULLETS

The original musket bullet was spherical lead ball smaller than bore, wrapper in a loosely fitted paper patch which served to hold the bullet in the barrel firmly upon the firing risked causing the barrel to explode, with the condition known as short start. The spherical bullet fitting ball to take the rifling grooves was more difficult to load, particularly when the bore of barrel was fouled from previous firings. For this reason, early rifles were not generally used for military purposes.

Fig.4.2 Spherical Bullets

4.3. GUN CHRONOGRAPH

A gun chronograph is an instrument to measure velocity of the projectile of the bullet fired by a gun/compressor. The ballistic precision chronograph has accuracy and calibrated among other chronographs. The chronograph consists of two sensing areas called chronograph screens, which contains projectile sensors that detect the passage of the bullet. The bullet is fired so it passes through both screens, the time it takes the bullet to travel the distance between the screens is measured electronically.

A shooting chronograph is one of the best tools to determine whether the gun is shooting properly or not. It can be used with variety of guns including paint balls, air soft guns, shot guns and rifles. The best chronograph needs to be very accurate and easy to use without breaking the bank.

Fig.4.3 Gun Chronograph

4.4 DOUBLE BENDING BEAM LOAD CELL

A load cell is used to convert the mechanical force in to electrical signals which will directly transfer the signals into reading which is directly proportional to force. Load cells are commonly used in weight industrial environment.

Load cell works on the principal of deformation of the strain gauge load cells, but the basic piezoelectric material proportional to the deformation of load cell generates a voltage output. They are installed on hoppers, reactors and allowed to control the weight in capacity. Ranges are specified for its ranged application. The senses of deformation by strain gauge electrical resistance change on obtained force.

4.5 HX711 AMPLIFIER

The HX711 load cell amplifier is used to get measurable data from load cell and strain gauge. It is small break out board for the HX711 IC that allows you to read load cell measurements by connecting the amplifier to your micro controller.

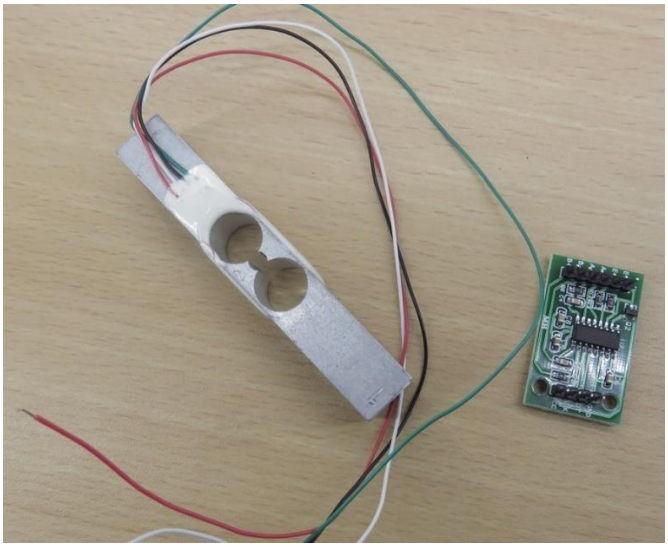

Fig.4.4 Load cell with HX711 amplifier.

4.6 ARDUINO PROGRAM

```
#include "HX711.h"

// HX711 circuit wiring constint

LOADCELL_DOUT_PIN = 2; constint

LOADCELL_SCK_PIN = 3; HX711

 scale; void setup()
{

Serial.begin(57600);

scale.begin(LOADCELL_DOUT_PIN, LOADCELL_SCK_PIN);

} void loop() { if
(scale.is_ready()) { long
reading = scale.read();
Serial.print("HX711 reading: ");

Serial.println(reading);

  } else

{

Serial.println("HX711 not found.");

  }

delay(1000);

}
```

4.7 PLATES AFTER BALLISTIC IMPACT TEST

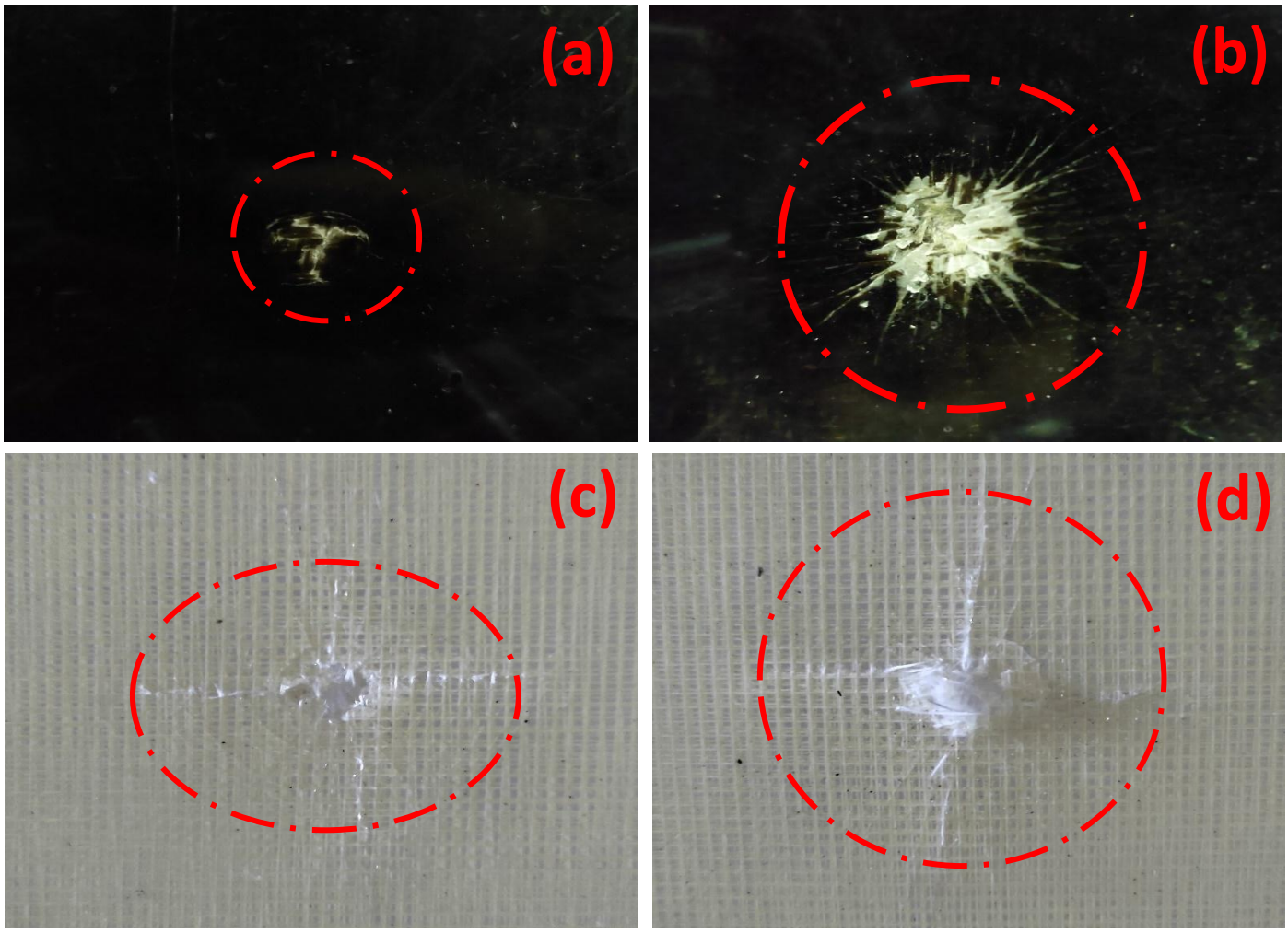

Fig.4.5 Pure Composites After Impact

B – BASALT FIBER

P – POLYPROPYLENE MESH

a – Basalt composite front face

b – Basalt composite back face

c – Polypropylene mesh composite front face

d – Polypropylene mesh composite back face

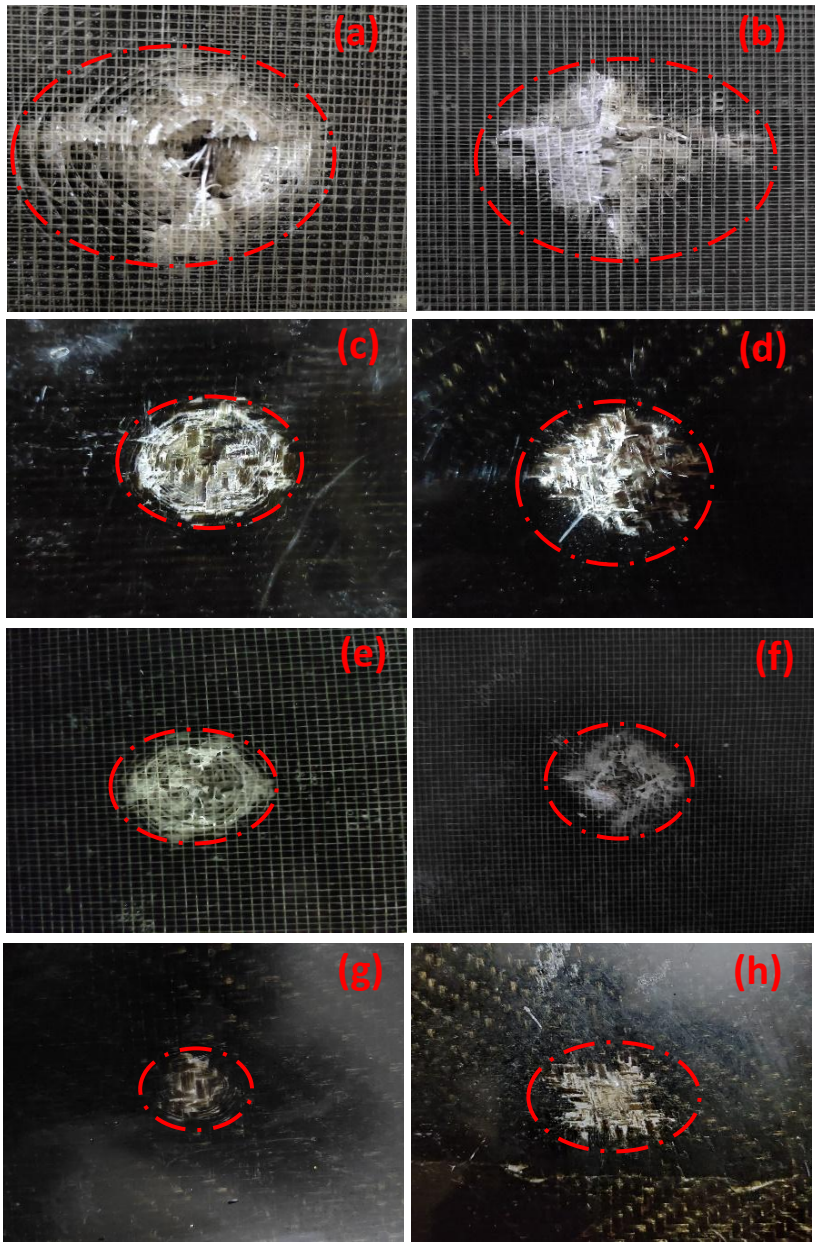

Fig.4.6 Hybrid Composites After Impact

a - PPBPP hybrid composite front face.

b - PPBPP hybrid composite back face.

c - BPPPB hybrid composite front face.

d - BPPPB hybrid composite back face.

e - PBBBP hybrid composite front face.

f - PBBBP hybrid composite back face.

g - BPBPB hybrid composite front face.

h - BPBPB hybrid composite back face.

CHAPTER 5

CALCULATIONS AND RESULTS

5.1 CALCULATIONS

5.1.1 SPECIFICATION OF BULLET

Material : Structural Steel

Geometry : Spherical

Diameter : 1.90 cm

5.1.2 SPECIFICATION OF FRAME

Total length between target and Pressure gun = 265 cm

Pipe length = 100 cm

Distance travelled by bullet in atmosphere = 165 cm

5.1.3 IMPACT ENERGY OF BULLET

Mass of bullet (m) = 27×10^{-3} g

Initial velocity (v_i) = 102 m/s

Distance (S) = 1.65 m

$W = 1/2\times (m.\ v^2)$ J

$= 1/2 \times (0.027\times102^2)$ J

$= 140.45$ J

MOMENTUM OF BULLET (Kg m/s)

Momentum (P) = m × v Kg m/s

P = 0.027×102

= 2.574 Kg m/s

ENERGY ABSORBED BY THE PLATES (J)

Energy absorbed (E) = $1/2 \times m\ (v_i^2 - v_r^2)$ J

Where m = Mass of the bullet

v_i = Initial velocity of the bullet (constant for all)

v_r = Residual velocity of the bullet

Basalt/epoxy Composite

v_r = 40.11 m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 40.11^2)$ J

= 118.74 J

PP Mesh/epoxy Composite

v_r = 47.41 m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 47.41^2)$ J

= 110.11 J

BP 1 Composite

v_r = 59.21 m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 59.21^2)$ J

= 93.13 J

BP 2 Composite

$v_r = 68.03$ m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 68.03^2)$ J

= 77.97 J

BP 3 Composite

$v_r = 44.93$ m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 44.93^2)$ J

= 113.20 J

BP 4 Composite

$v_r = 36.27$ m/s

Energy absorbed (E) = $1/2 \times 0.027\ (102^2 - 36.27^2)$ J

= 112.69 J

5.2 RESULTS

The below graph depicts the load vs time chart for each of the specified samples and the area under each curve gives the energy absorption factor for each of the specimen.

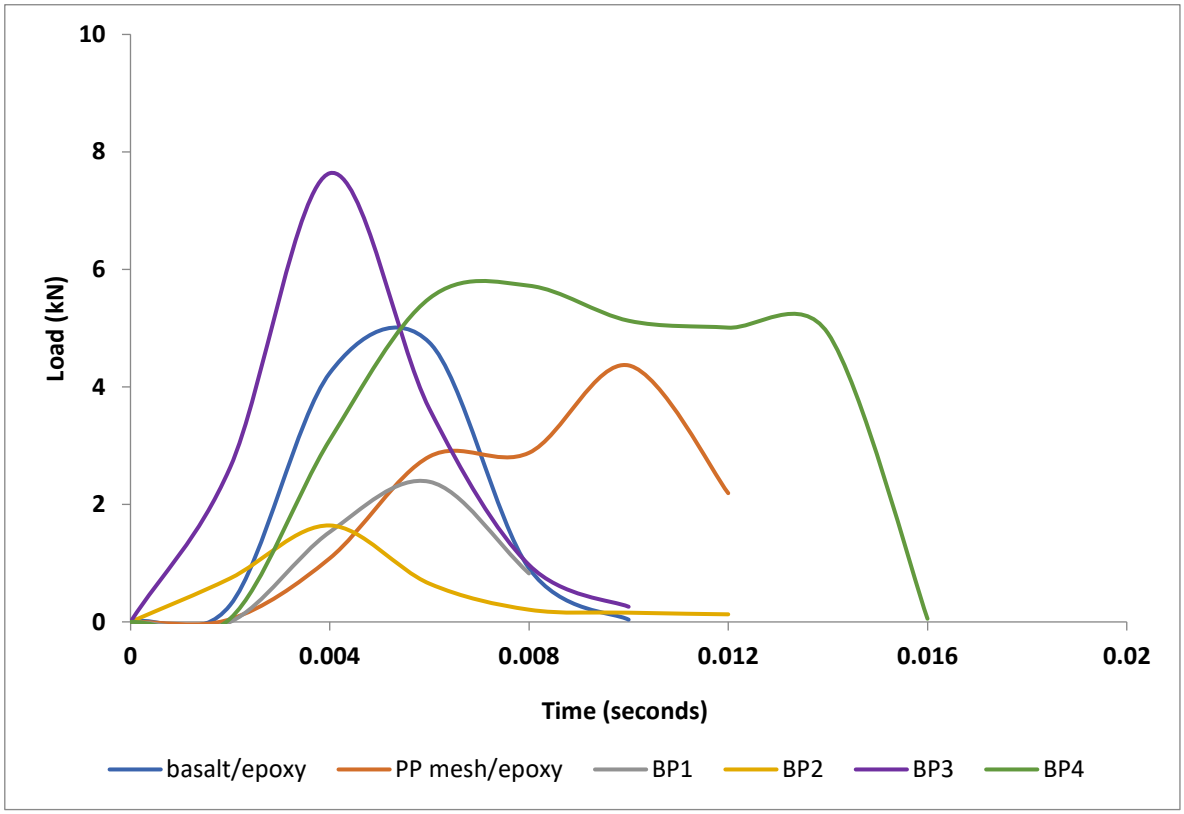

Fig.5.1 Time v/s Load

BP1 – PPBPP Hybrid composite.

BP2 - BPPPB Hybrid composite.

BP3 - PBBBP Hybrid composite.

BP4 - BPBPB Hybrid composite.

The peak load for each of the specimen is plotted as a bar graph and the peak load values were taken in kN.

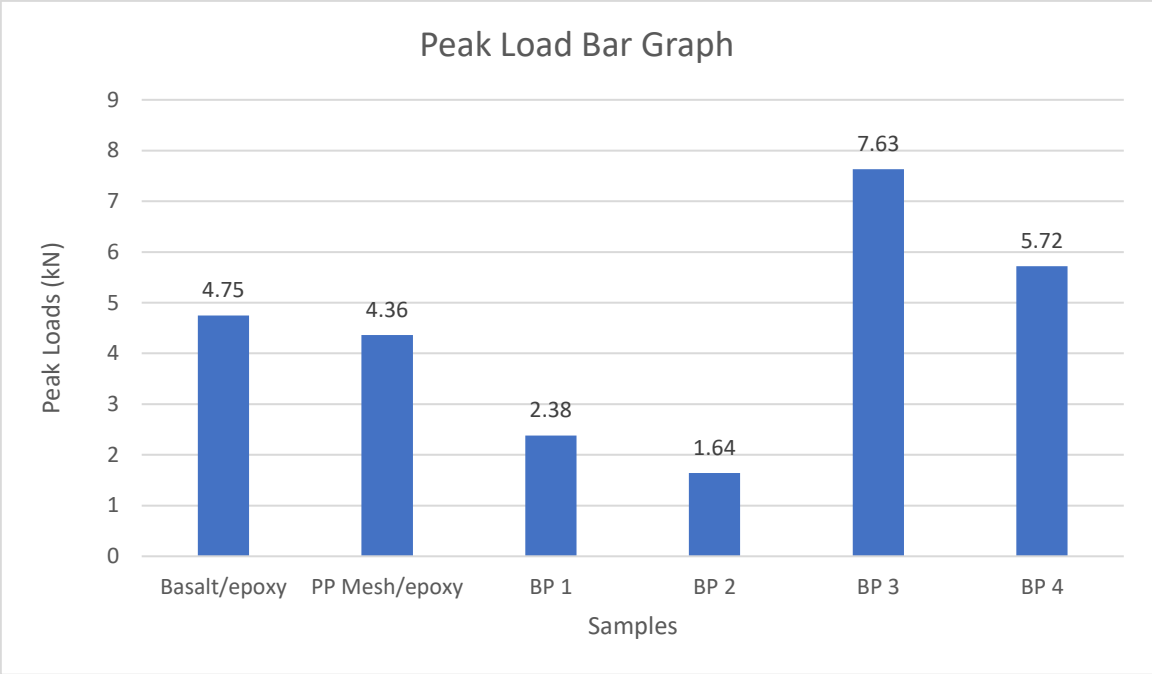

Fig.5.2 Peak Loads

From the above peak loads bar graph, It is observed that different samples have different values of load and sample 5(BP3) hybrid composite has the highest peak load followed by sample 6(BP4), sample 1(Basalt/epoxy), sample 2(PP Mesh/epoxy), sample 3(BP1) & sample 4(BP2).

5.3 DAMAGE AREA

The below tables are the bar graphs for the damage area occurred in front & back face of the pure and hybrid composites.

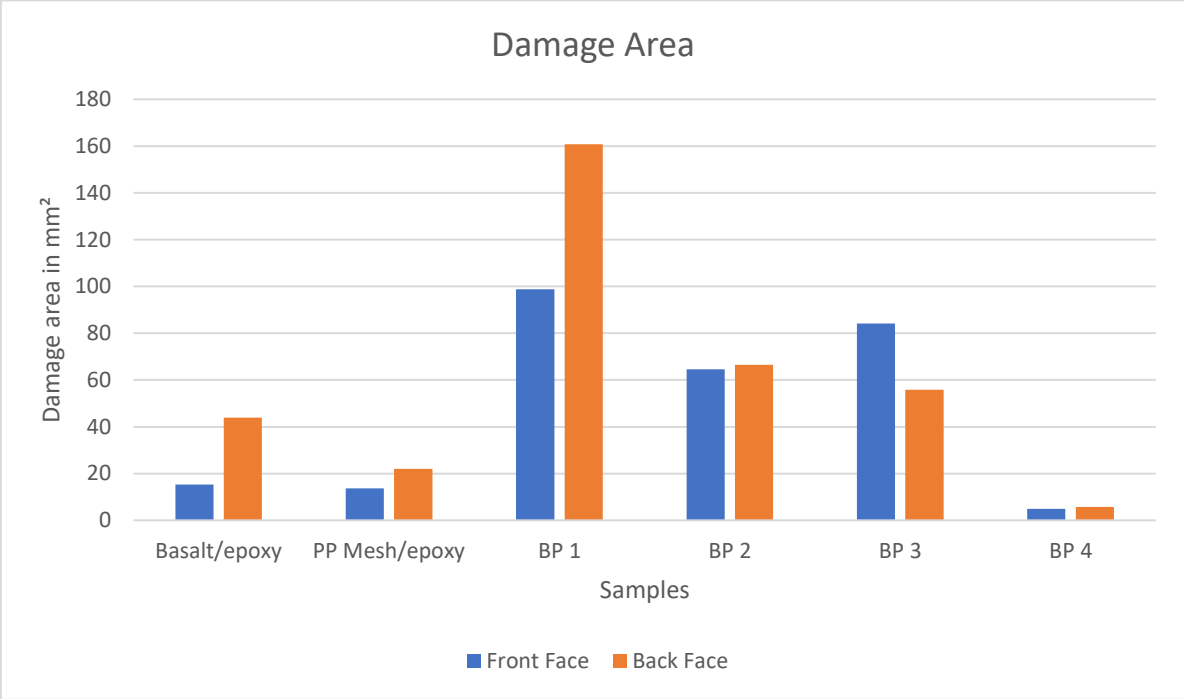

Fig.5.4 Damage Area

The deductions drawn from the above graph were that the maximum area of damage was observed in BP1 and the minimum area of damage was observed in BP4 and the most efficiency in the area of damage and energy absorption was obtained in BP2.

Table.5.3 Result Compilation

Composite	**Impact Energy (J)**	**Residual velocity (m/s)**	**Energy absorbed (J)**
Basalt/epoxy	140.54	40.11	118.74
PP mesh/epoxy	140.54	47.41	110.11
BP1	140.54	59.21	93.13
BP2	140.54	68.03	77.97
BP3	140.54	44.93	113.20
BP4	140.54	36.27	122.69

The calculations done after the tests were compiled by each specimen in order and were tabulated in the above table. According to the test results the maximum energy absorption was observed in the sample BP4 followed by pure basalt and the least energy absorption was observed in BP2 sample.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this current project six different samples made of basalt/polypropylene mesh were made using six different stacking sequences to be analysed for impact response.

The conclusions made from the current work were that the hybrid composites possess better energy absorption characteristics and superior load bearing abilities compared to pure composites.

The maximum energy absorption was observed in alternately stacked basalt and poly-propylene plate i.e. BP4 followed by pure basalt plate and the least energy absorption was given by BP2 sample.

The maximum area of damage was seen in BP1 sample while the least amount of area damage was received by BP4.

6.2 FUTURE WORK

The above discussed work can be extended in the following ways.

- The damage assessment to be done through non-destructive testing.
- The energy absorption and the damage area assessment to be done at a higher impact velocity than 102m/s.

CHAPTER 7

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